Examiner only

6. A student says that a simple pendulum oscillates with simple harmonic motion with a period, T, of:

$$T = 2\pi\sqrt{\frac{l}{g}}$$

where l is the length of the pendulum and g is the acceleration due to gravity.

(a) Describe what is meant by a simple pendulum. [1]

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(b) State what is meant by simple harmonic motion. [2]

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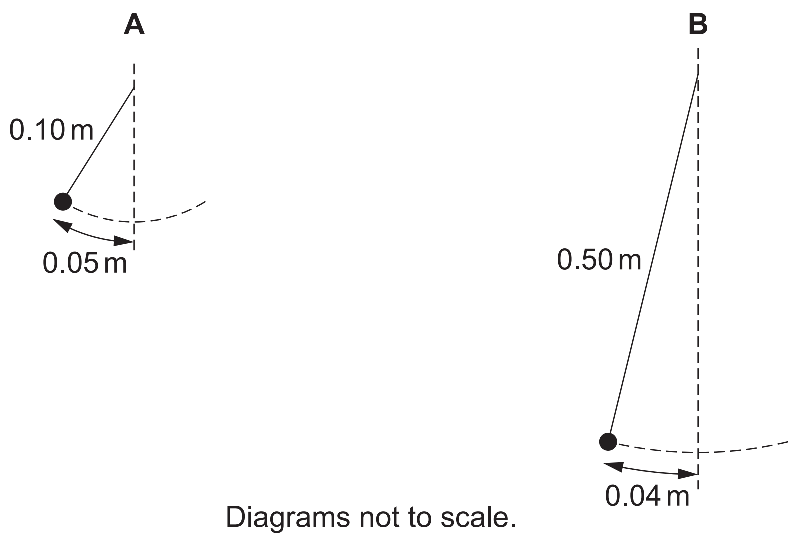
(c) The student sets up two experimental systems to investigate simple harmonic motion. System A is a pendulum of length 0.10 m oscillating with an amplitude of 0.05 m. System B is a pendulum of length 0.50 m oscillating with an amplitude of 0.04 m. Give two reasons why system B is better than system A to investigate simple harmonic motion. [2]



Diagrams not to scale.

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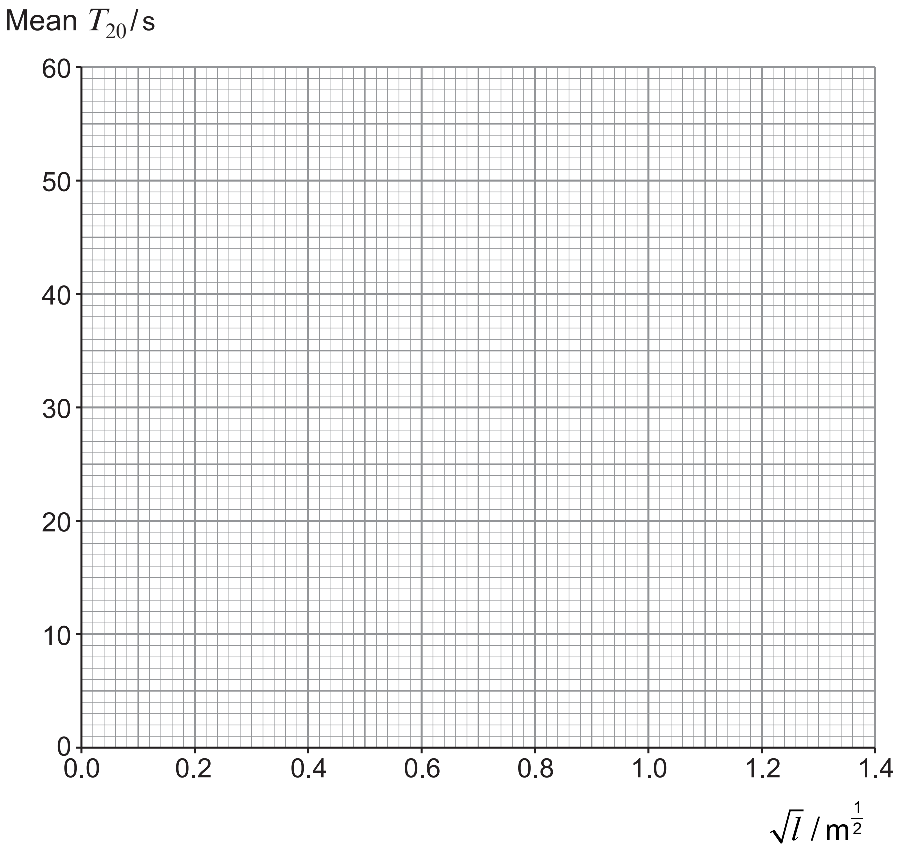
(d) Measurements for an oscillating simple pendulum are given in the table below, where l is the length of the pendulum and T_{20} is the time taken for 20 oscillations. For each value of l four values of T_{20} are measured.

l / m	$\sqrt{l}$ / $m^{\frac{1}{2}}$	T_{20} / s	Mean T_{20} / s	Uncertainty T_{20} / s
0.250	0.50	19, 21, 22, 20	21	2
0.500	0.71	26, 29, 26, 29	28	2
0.750	....	35, 34, 33, 36	....	....
1.000	....	42, 39, 40, 41	....	....
1.250	....	45, 42, 44, 45	....	....
1.500	....	48, 50, 49, 52	....	....

(i) **Complete the table** by determining $\sqrt{l}$, the mean time for 20 oscillations, T_{20} and the uncertainty in T_{20}. Values have been inserted for the first two lengths. [3]

(ii) Plot mean T_{20} against $\sqrt{l}$ on the grid below. Include error bars on the T_{20} axis only. Draw a line of maximum gradient and a line of minimum gradient. [4]



Examiner only

(iii) Justify why error bars for $\sqrt{l}$ are not plotted on the graph. [2]

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(iv) Use the maximum and minimum gradients to determine a mean value for the acceleration due to gravity, g, and its **percentage** uncertainty. [5]

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(e) Another student says that the gradient of the graph would be approximately twice as large if the experiment was carried out on the Moon, where the acceleration is approximately $0.2g$. Investigate if this statement is correct, justifying your answer. [3]

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